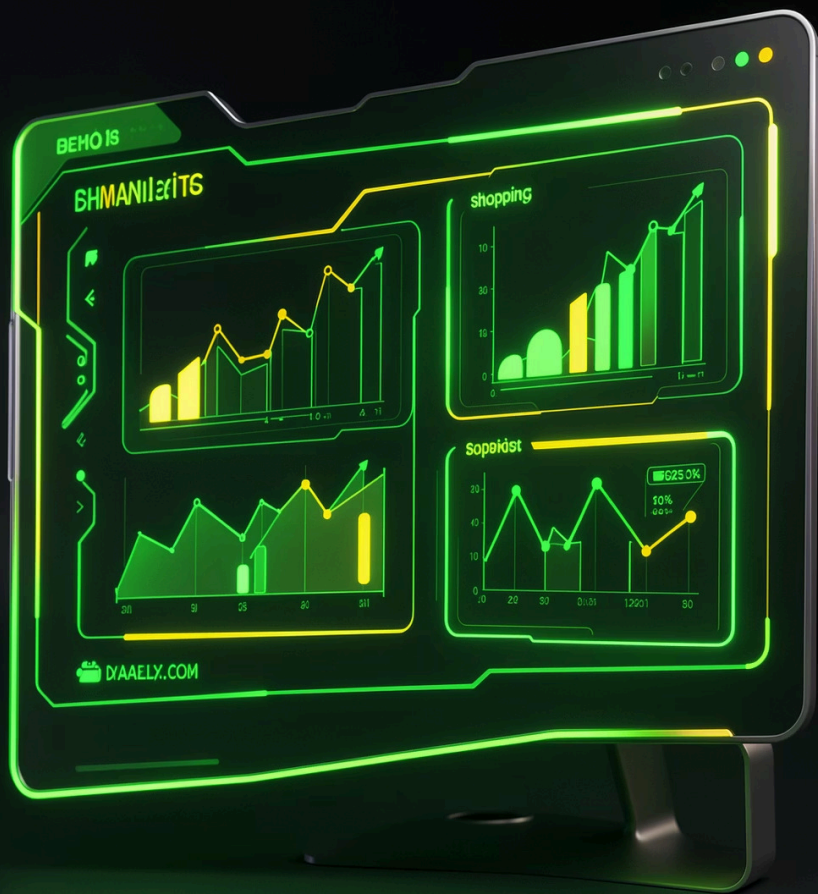




Customer Shopping Behavior Analysis

A deep-dive into transactional data from **3,900 purchases** across multiple product categories — uncovering spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

DATA ANALYTICS

RETAIL INSIGHTS

PYTHON · SQL · POWERBI

Dataset at a Glance

The dataset contains **3,900 rows** and **18 columns**, covering three core feature groups. One data quality issue was identified: **37 missing values** in the Review Rating column, later resolved during cleaning.

Customer Demographics

- Age, Gender, Location
- Subscription Status

Purchase Details

- Item, Category, Amount
- Season, Size, Color

Shopping Behavior

- Discounts, Promo Codes
- Frequency, Ratings, Shipping

Data Preparation in Python

The dataset was loaded and cleaned using **pandas**. Key steps included using `df.info()` and `.describe()` for initial exploration, imputing missing Review Ratings with the **median per product category**, and renaming columns to snake_case for consistency.

01

Feature Engineering

Created `age_group` by binning ages and `purchase_frequency_days` from purchase data.

03

Database Integration

Connected Python to **PostgreSQL** and loaded the cleaned DataFrame for SQL analysis.

02

Data Consistency

Verified `discount_applied` and `promo_code_used` were redundant — dropped `promo_code_used`.



SQL Analysis: Revenue & Spending Insights

Structured queries in **PostgreSQL** answered key business questions around revenue, discounts, and shipping behavior.

Revenue by Gender

Compared total revenue generated across gender segments to identify high-value audiences.

High-Spend Discount Users

Identified customers who used discounts yet still spent **above the average** purchase amount.

Shipping Type Comparison

Compared average purchase amounts between **Standard** and **Express** shipping — express users showed higher average spend.

Subscribers vs. Non-Subscribers

Subscribers demonstrated higher average spend and contributed greater total revenue compared to non-subscribers.

SQL Analysis: Products, Ratings & Loyalty

1

Top 5 Products by Rating

Identified products with the highest average review ratings across all categories.

2

Discount-Dependent Products

Found the 5 products with the **highest percentage** of discounted purchases — flagging margin risk.

3

Top 3 Products per Category

Listed the most purchased products within each product category for assortment planning.

4

Repeat Buyers & Subscriptions

Customers with **more than 5 previous purchases** were analyzed for their likelihood to subscribe.

Customer Segmentation

Customers were classified into three segments based on purchase history, enabling targeted engagement strategies for each group.



New Customers

First-time buyers with limited purchase history. Priority: onboarding and first-repeat conversion.



Returning Customers

Moderate purchase history. Priority: deepen engagement and move toward loyalty status.

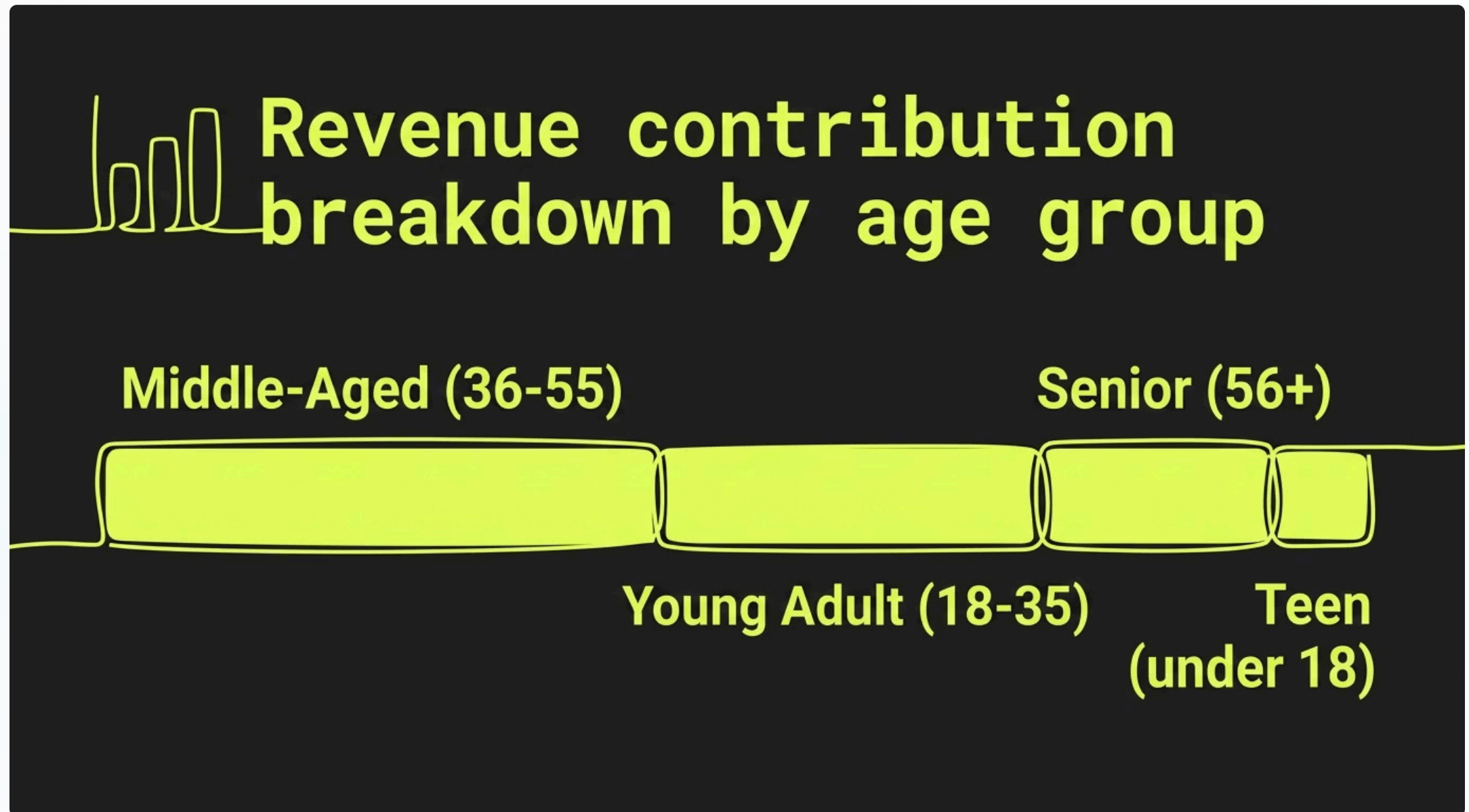


Loyal Customers

High purchase frequency. Priority: retention, rewards, and subscription conversion.

Revenue Contribution by Age Group

SQL analysis revealed how different age groups contribute to overall revenue — a key input for targeting marketing spend and personalizing offers across the customer lifecycle.



 Age group segmentation was created via feature engineering — binning the raw Age column into meaningful cohorts for analysis.



PowerBI Dashboard

All findings were consolidated into an interactive **PowerBI dashboard**, providing stakeholders with a single view of customer behavior, revenue trends, segment performance, and product insights — enabling real-time, data-driven decision-making.

Business Recommendations

Drive Subscription Growth

Promote exclusive subscriber benefits and personalized offers to increase conversions and retention.

Strengthen Customer Loyalty

Implement a structured rewards program to encourage repeat purchases and transition customers into high-value loyalty segments.

Optimize Discount Strategy

Reassess discount policies to balance revenue growth with margin protection and long-term profitability.

Enhance Product Positioning

Prioritize top-rated and best-selling products in marketing campaigns to maximize visibility and conversion rates.

Refine Targeted Marketing

Focus campaigns on high-revenue segments — key age groups and express shipping users — to improve ROI.



Key Takeaways

3,900 Transactions Analyzed

Across 18 features covering demographics, purchases, and behavior.

Full Analytics Pipeline

End-to-end workflow: Python cleaning → PostgreSQL analysis → PowerBI visualization.

Actionable Insights

5 strategic recommendations to grow subscriptions, loyalty, and revenue.